Assignment-1

To build and visualize a smart cleaning agent in Python

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1 Introduction

Artificial Intelligence agents are autonomous entities that perceive their environment through sensors and act upon that environment using actuators. One of the most fundamental examples used to explain intelligent agents is the **Vacuum Cleaner Agent**.

This assignment demonstrates the implementation and visualization of a **simple reflex agent** operating in a **2×2 grid environment**. The agent observes the cleanliness of the current room and performs actions based on predefined rules.

2 Objective

The objectives of this assignment are:

- To design a reflex-based vacuum cleaner agent
- To simulate agent behavior in a multi-room environment
- To visualize agent movement and cleaning actions
- To analyze the agent's decision-making process step by step

3 Environment Description

The environment consists of four rooms arranged in a 2×2 grid:

- Room1 (Top-Left)
- Room2 (Top-Right)
- Room3 (Bottom-Left)
- Room4 (Bottom-Right)

Initially:

- Room2 is **Dirty**
- All other rooms are **Clean**

The agent starts in **Room1**.

4 Agent Description

The vacuum cleaner agent is a **Simple Reflex Agent**. Its behavior is governed by the following rules:

- If the current room is **Dirty** → **Clean the room**
- If the current room is **Clean** → **Move to the next room**

The agent moves sequentially through the rooms in a cyclic manner.

5 Execution and Results

5.1 Step 1

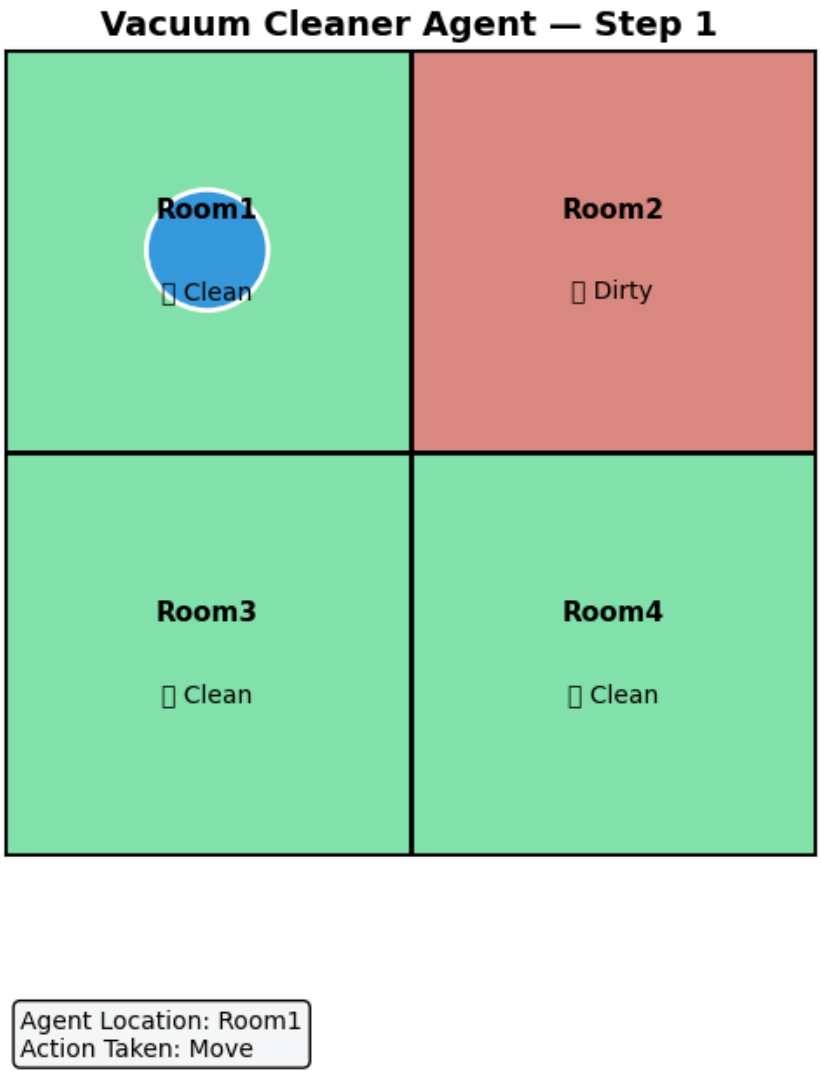

Figure 1: Step 1: Agent starts in Room1

The agent begins in Room1, which is already clean. Since no cleaning is required, the agent decides to **move** to the next room.

5.2 Step 2

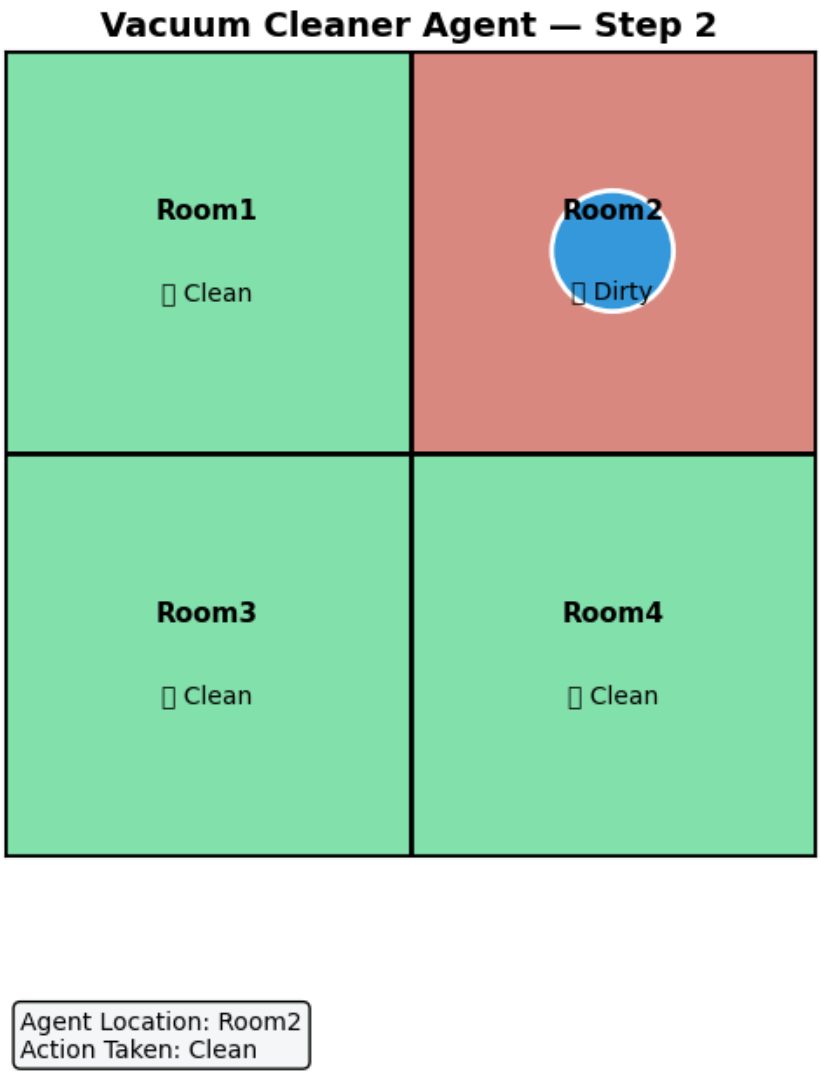

Figure 2: Step 2: Agent reaches dirty Room2

The agent enters Room2 and detects dirt. According to the reflex rule, the agent performs the **Clean** action.

5.3 Step 3

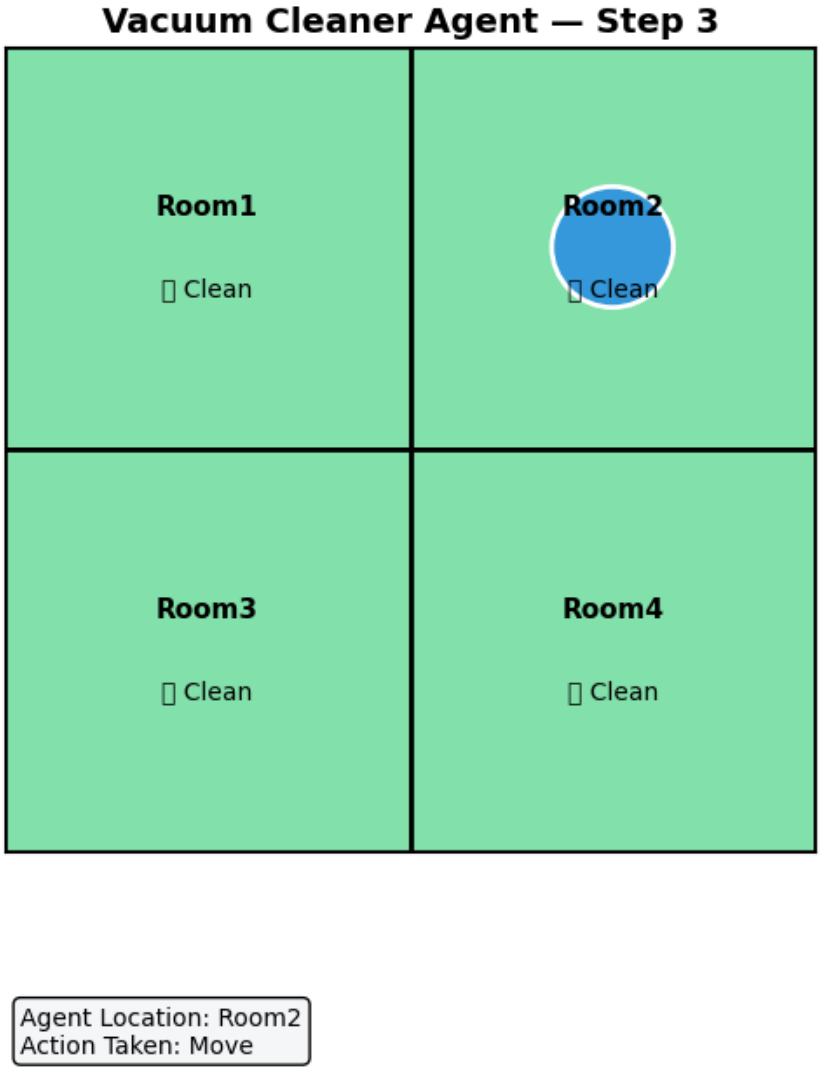

Figure 3: Step 3: Room2 cleaned

Room2 is now clean. The agent has no cleaning task and decides to **move** to the next room.

5.4 Step 4

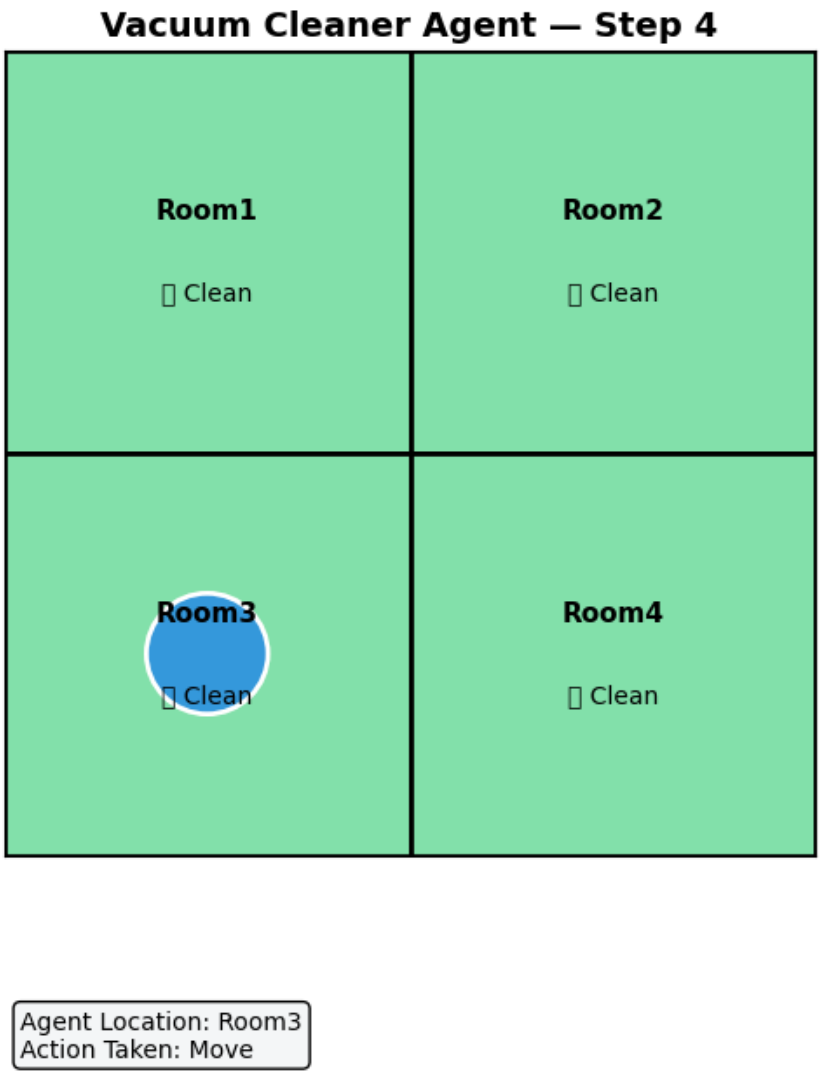

Figure 4: Step 4: Agent moves to Room3

The agent reaches Room3, which is clean. The agent performs no cleaning and continues moving.

5.5 Step 5

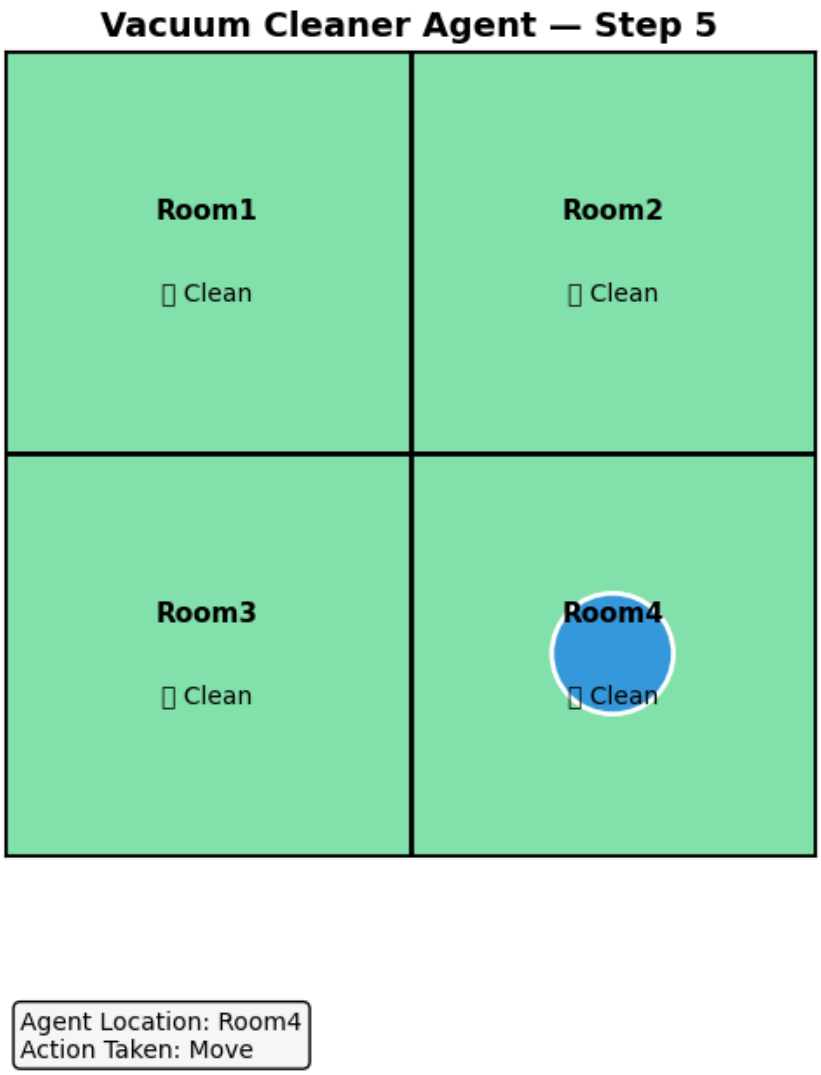

Figure 5: Step 5: Agent moves to Room4

Room4 is also clean. The agent again decides to move forward.

5.6 Step 6

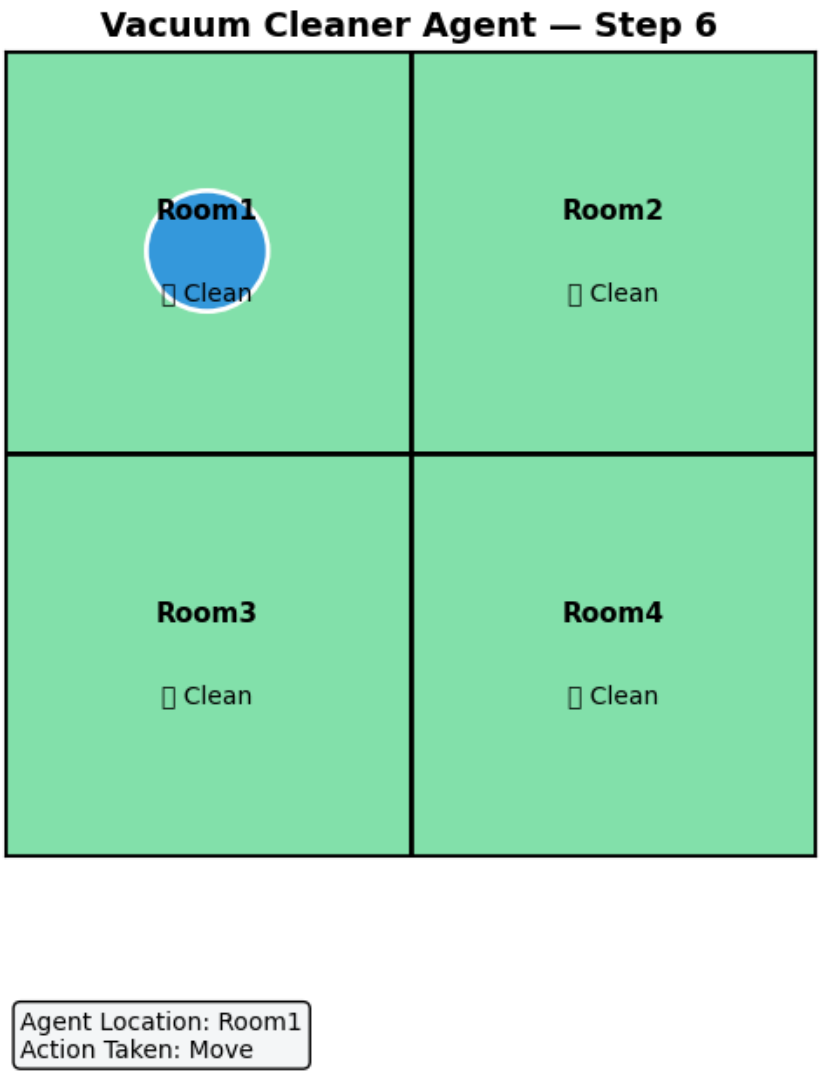

Figure 6: Step 6: Agent returns to Room1

The agent cycles back to Room1. Since the room remains clean, the agent moves again.

5.7 Step 7

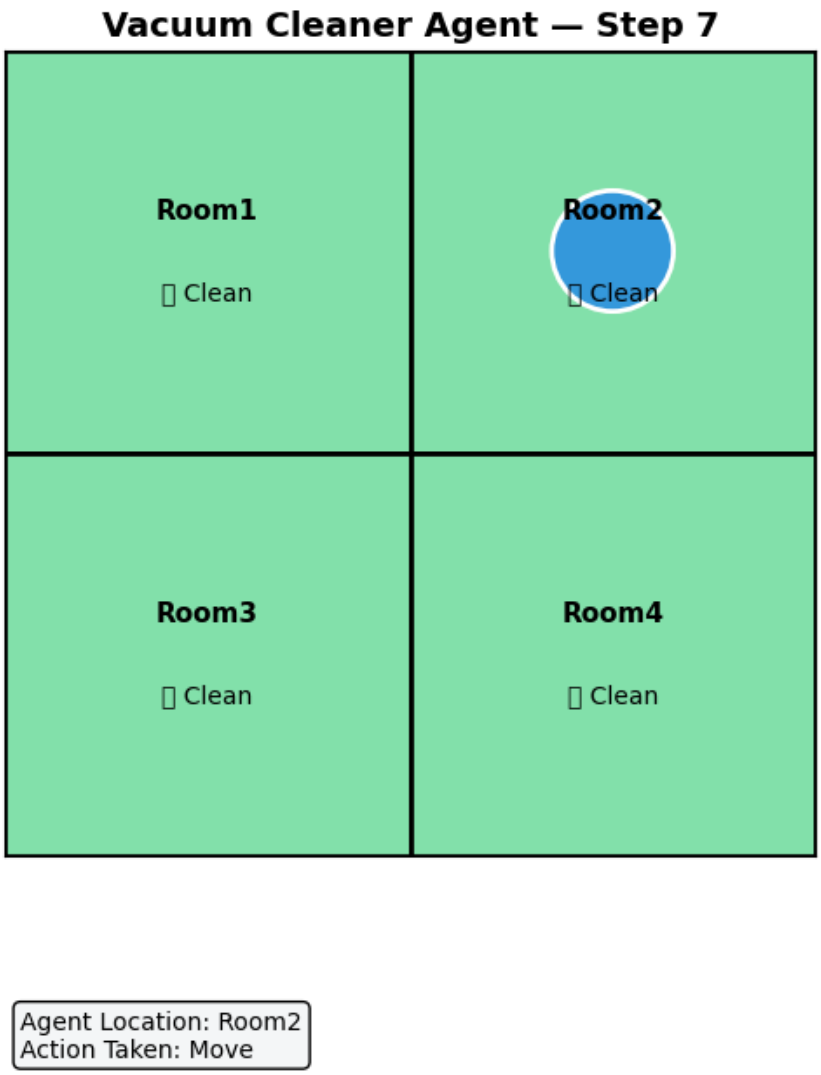

Figure 7: Step 7: Agent revisits Room2

Room2 is still clean from the earlier cleaning action. The agent confirms cleanliness and proceeds.

5.8 Step 8

Figure 8: Step 8: Agent moves to Room3

The agent continues its cyclic movement, maintaining a clean environment.

6 Discussion

This simulation demonstrates that a simple reflex agent:

- Can effectively clean a deterministic environment
- Reacts only to the current percept
- Does not require memory or planning

However, such agents are limited in more complex or unpredictable environments.

7 Conclusion

The vacuum cleaner agent successfully cleaned the environment and maintained cleanliness using simple condition-action rules. This experiment highlights the working of reflex agents and serves as a foundation for understanding more advanced intelligent agents such as model-based, goal-based, and learning agents.

8 Project GitHub Repository

The complete source code and implementation of this assignment are available at the following GitHub repository:

`https://github.com/Prathamesh-Palatshaha/MTech-27/tree/main/AI-624-Agentic-AI/ASSIGNMENT%201`